

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

Term 2 Assignment Answer Book

NAME: Marking guideline

CLASS: 12 _____

TEACHER: _____

	Mark				
Question	H	T	U	Marker	HOD
1					
2					
3					
TOTAL (Out of 50)					
%					

for 1.1.1 learner uses incorrect formula then no marks.
 If learner uses correct formula but makes a subs error then can award ✓A for answer.

QUESTION 1

	Solution	Marks
1.1.1	$1 + i_{eff} = \left(1 + \frac{i_{nom}}{m}\right)^m$ $1 + i_{eff} = \left(1 + \frac{15\%}{12}\right)^{12} \checkmark A$ $i_{eff} \approx 16,08\% \checkmark$	(2)
1.1.2	$F = \frac{x[(1+i)^n - 1]}{i}$ $= \frac{7000 \left[\left(1 + \frac{15\%}{12}\right)^{48} - 1 \right]}{\frac{15\%}{12}} \checkmark A \text{ Subs into correct formula}$ $= R456\,598,72 \checkmark$	(2)
1.1.3	$1\,000\,000 = \frac{7000 \left[\left(1 + \frac{15\%}{12}\right)^n - 1 \right]}{\frac{15\%}{12}} \checkmark \text{subs into correct formula}$ $\frac{39}{14} = \left(\frac{81}{80}\right)^n$ $n = \log_{\frac{81}{80}} \left(\frac{39}{14}\right) \checkmark \text{Use of log (not required to show)}$ $n = 82,4715... \checkmark$ $\therefore 83 \text{ months } \checkmark CA$	(4)

	Solution	Marks
1.2.1	$A = P(1+i)^n$ $= 45000 \left(1 + \frac{9\%}{4}\right)^3 \checkmark A$ $= 48106,36$ $P = \frac{x \left[1 - (1+i)^{-n}\right]}{i}$ $\overset{CA}{\checkmark} A = P \quad \checkmark n=13$ $48106,36 = \frac{x \left[1 - \left(1 + \frac{9\%}{4}\right)^{-13}\right]}{\frac{9\%}{4}}$ $\therefore x = 4309,22 \checkmark CA$	(4)
1.2.2	$\text{Total repayments} = 4309,22 \times 13$ $= 56019,86 \checkmark CA \text{ 'x'}$ $\text{Interest} = 56019,86 - 45000$ $= 11019,86 \checkmark$	(2)

	Solution	Marks
1.2.3	$OB = A(1+i)^m - \frac{x[(1+i)^m - 1]}{\frac{i}{4}}$ $= 45000 \left(1 + \frac{9\%}{4}\right)^9 - \frac{4309,22 \left[\left(1 + \frac{9\%}{4}\right)^6 - 1\right]}{\frac{9\%}{4}}$ $= 27\,623,11 \checkmark$	(4)
	Alternative:	[18]

$$OB = \frac{x[1 - (1+i)^{-n}]}{\frac{i}{4}}$$

$$= \frac{4309,22 \left[1 - \left(1 + \frac{9\%}{4}\right)^{-7}\right]}{\frac{9\%}{4}}$$

$$\approx 27\,623,16 \checkmark$$

(4)

Question 1.2

If learner does not defer annuity but answers as follows:

2.1. $45\,000 = \frac{x \left[1 - \left(1 + \frac{9\%}{4} \right)^{-16} \right]}{\frac{9\%}{4}}$ ✓ Use P
i must be $\frac{9\%}{4}$

$\therefore x = 3380,25$ ✓ CA max $\frac{2}{4}$

2.2. Total repayments = $3380,25 \times 16$
 $= 54084$

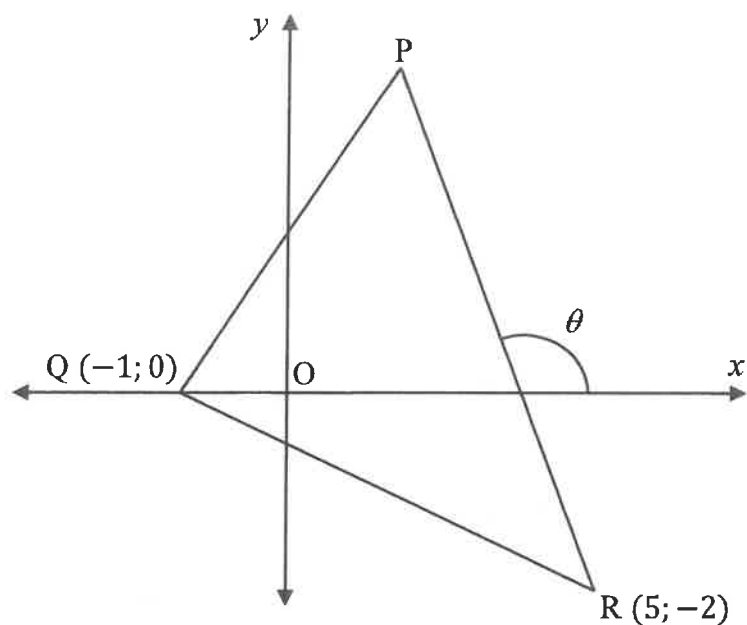
Interest = 9084 ✓ CA max $\frac{1}{2}$

2.3. $OB = 45000 \left(1 + \frac{9\%}{4} \right)^6 - \frac{3380,25 \left[\left(1 + \frac{9\%}{4} \right)^6 - 1 \right]}{\frac{9\%}{4}}$ ✓ m

$= 29970,00$ ✓ CA

max $\frac{2}{4}$

QUESTION 2

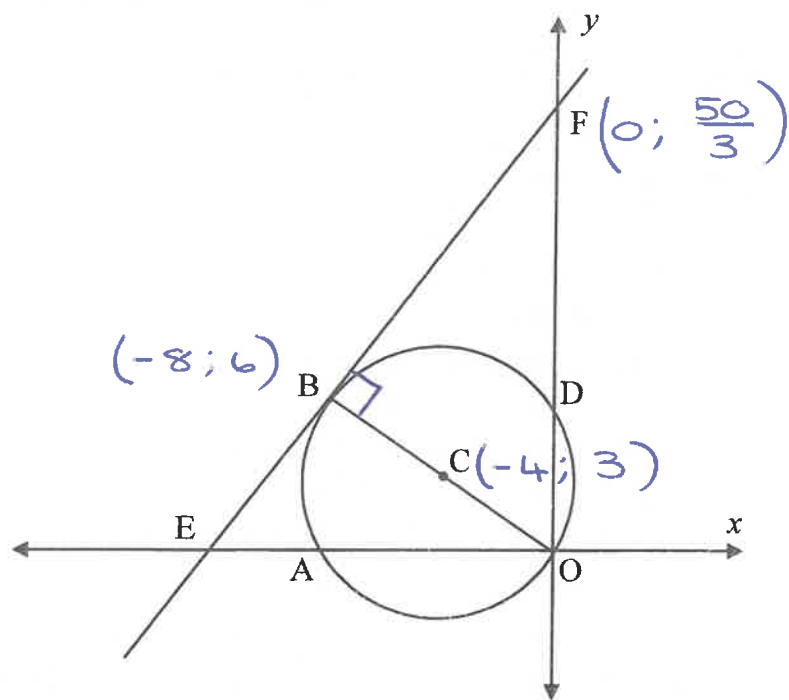


	Solution	Marks
2.1	$QR = \sqrt{(-1-5)^2 + (0+2)^2} \checkmark$ $= 2\sqrt{10} \checkmark$	(2)
2.2	$PQ: y = 3x + 3 \quad m_{PQ} = 3 \checkmark$ $m_{QR} = \frac{0+2}{-1-5} = -\frac{1}{3} \checkmark$ $m_{PQ} \times m_{QR} = 3 \times -\frac{1}{3} = -1 \checkmark$ $\therefore \hat{PQR} = 90^\circ$	(3)

	Solution	Marks
2.3	$m_{PR} = -2$	(2)
	$\tan \theta = -2 \checkmark^m \tan \theta = m$	
	$\theta = 116,57^\circ \checkmark \text{CA if } m_{PR} \text{ incorrect}$	
2.4	$3x + 3 = -2x + 8 \checkmark^m \text{Point of intersection}$	(3)
	$5x = 5$	
	$\therefore x = 1 \checkmark$	
	$\therefore y = 6$	
	$P(1; 6)$	
	$\checkmark \text{Must be in coordinates}$	

	Solution	Marks
2.5	PR is diameter Chord subtends 90° $M_{PR} \left(\frac{1+5}{2}, \frac{6-2}{2} \right)$ Centre $(3; 2)$ $(x-3)^2 + (y-2)^2 = r^2$ $(5-3)^2 + (-2-2)^2 = 20$ $\therefore (x-3)^2 + (y-2)^2 = 20$	(4)
2.6	Centre $(x; -2x+8)$ $Q(-1; 0)$ $S(8; 3)$ $CQ^2 = CS^2$ $(-1-x)^2 + (0+2x-8)^2 = (8-x)^2 + (3+2x-8)^2$ $1+2x+x^2 + (2x-8)^2 = 64-16x+x^2 + (2x-5)^2$ $1+2x+x^2 + 4x^2 - 32x + 64 = 64 - 16x + x^2 + 4x^2 - 20x + 25$ $-30x + 1 = -36x + 25$ $6x = 24$ $x = 4$ $\therefore y = 0$ Centre $(4; 0)$ must be in coordinates	(5)
		[19]

QUESTION 3



	Solution	Marks
3.1	$x^2 + 8x + (4)^2 + y^2 - 6y + (-3)^2 = 0 + 16 + 9$ $(x+4)^2 + (y-3)^2 = 25$ $\therefore C(-4; 3)$	completing the square (2)
3.2	$B(-8; 6)$	(2)

	Solution	Marks
3.3	$m_{\text{rad}} = \frac{0-6}{0+8} = -\frac{3}{4} \checkmark$ $\therefore m_{\text{tan}} = \frac{4}{3} \text{ } \checkmark_R \text{ (rad } \perp \text{ tan)}$ $y-6 = \frac{4}{3}(x+8) \checkmark_m$ $y = \frac{4x}{3} + \frac{50}{3} \checkmark$	(4)
3.4	$BO = 10 \text{ units } \checkmark_A$ $\text{Area } \triangle OBF = \frac{1}{2} BO \cdot BF \checkmark_m$ $= \frac{1}{2} (10) \left(\frac{40}{3} \right)$ $= \frac{200}{3} \text{ units}^2$ Accept either $\approx 66,67 \text{ units}^2 \checkmark$	(3)
3.5	$K = -5 \text{ } \checkmark_A \text{ or } k = 5 \text{ } \checkmark_A$	(2)
		[13]